

OPENALEX SURVEY · AUTONOMOUS AGENTS

Are your isolated AI agents actually holding your systems back?

1 Unified Framework



Arunachalam Venkatachalapathy

AI Agent & Forward Deployment Engineer

THE TENSION

Most CleanTech engineers assume training agents in isolated, hyper-specific environments is the safest path to optimization. It isn't. It leaves them fragile and unable to make human-like decisions when real-world telemetry shifts.

INSIGHT 2

**Leverage Web-Scale Knowledge.
Stop restricting agents to closed
datasets. Modern LLM agents use
vast web intelligence to navigate
complex, multi-variable ESG and
grid telemetry dynamically.**

INSIGHT 3

Deploy a Unified Framework.
Structure your deployments with four key pillars: profiling, memory, planning, and action. This framework mimics human cognitive workflows for resilient decision-making.

INSIGHT 4

**Apply Across Three Domains.
These agents transcend simple
automation. They are actively
orchestrating workflows across
social sciences, natural sciences,
and complex grid engineering.**

INSIGHT 5

The Automation Paradox. While we build agents to automate, we cannot automate their evaluation. Robust deployments require subjective human-in-the-loop testing, not just automated metric scores.

1 Unified Framework

1 unified framework now standardizes agent design across social, natural, and engineering sciences globally.

A SURVEY ON

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ACTION

Save this for your next AI agent deployment.

@arunachalamvenv

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